

One Interval, Two Cutoffs

AP Statistics · Unit 4 · Topic 4.3 · B: 2027-01-15 · E: 2027-03-03 · TEACHER KEY

CED TETHER

- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.D** — Justify a claim based on a confidence interval.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.S** — Interpret a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.B]
- **EK UNC-4.S.1** — A confidence interval for a population mean either contains the population mean or it does not, because each interval is based on data from a random sample, which varies from sample to sample.
- **EK UNC-4.S.2** — We are C% confident that the confidence interval for a population mean captures the population mean.
- **EK UNC-4.S.3** — An interpretation of a confidence interval for a population mean includes a reference to the sample taken and details about the population it represents.
- **LO UNC-4.T** — Justify a claim based on a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.D]
- **EK UNC-4.T.1** — A confidence interval for a population mean provides an interval of values that may provide sufficient evidence to support a particular claim in context.
- **LO UNC-4.U** — Identify the relationships between sample size, width of a confidence interval, confidence level, and margin of error for a population mean. [Skill 4.A]
- **EK UNC-4.U.1** — When all other things remain the same, the width of a confidence interval for a population mean tends to decrease as the sample size increases.
- **EK UNC-4.U.2** — For a single mean, the width of the interval is proportional to $1/\sqrt{n}$.
- **EK UNC-4.U.3** — For a given sample, the width of the confidence interval for a population mean increases as the confidence level increases.

Essential question: How can one confidence interval support a bounded claim while leaving a stronger claim unresolved?

DOK-3 skill: 4.D · **FRQ pattern:** mean-interval-two-cutoffs-and-scope

DOK rationale: Part (c) coordinates both endpoints, two cutoffs, random-sample scope, and the mean-versus-individual distinction to adjudicate two interpretations and trace an advertising consequence.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask which interval feature evaluates each cutoff.
- Begin the 7.3 video follow-along at minute 5.
- Return at minute 33. Require separate decisions for the two cutoffs and the individual-time claim.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one approach and cite an endpoint or the parameter.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- Why does 44.877 settle the 45-second claim but not the 42-second claim?
- Does this interval describe a population mean or 95 percent of individual transactions?
- Which population can the random sample represent?

Adult role

Solo teacher. Test each cutoff against the interval, then separate the mean from individual times.

[WARN] WATCH FOR

- Using only the lower endpoint to support a less-than claim.
- Treating 95 percent confidence as a probability that the fixed population mean is in this realized interval.
- Claiming that 95 percent of individual setup times fall between the confidence-interval endpoints.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-warranted-approach** (required) — Selects the full-interval, population-mean interpretation
- **separates-two-cutoffs** (required) — Uses 44.877 below 45 and 42 inside the interval
- **uses-whole-interval** (required) — Applies the entire-interval rule for a less-than claim
- **distinguishes-mean-from-individuals** (required) — Separates the population mean from individual times
- **traces-advertising-consequence** (required) — Names both unsupported advertising claims

E	Uses both cutoff locations, the whole-interval rule, the correct parameter, and both advertising consequences.
P	Evaluates both cutoffs but incompletely explains interval logic or the mean-versus-individual distinction.
I	Uses only values below a cutoff or treats the interval as covering 95 percent of individual times.

[STAR] TODAY'S PROBLEM — annotated

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE (5 min)

Which approach is more defensible? Cite an endpoint or identify the quantity being estimated.

Key: Not graded. Students should identify the parameter and an interval feature before deciding.

Rules callout “LET THE ENTIRE INTERVAL GOVERN THE CLAIM (keep on the page)” is printed on the student sheet.

DOK 1 (a) Define the population mean and locate 42 and 45 relative to the interval.

Key: Let μ be the mean setup time for all 800 kiosk transactions that month. The value 42 is inside (39.923, 44.877); 45 is above the upper endpoint.

DOK 2 (b) Verify random, 10 percent, and shape conditions. Reproduce the standard error, margin of error, and interval; interpret it in context.

Key: The SRS satisfies $25 \leq 0.10(800) = 80$. With a symmetric, outlier-free sample and $n < 30$, the t shape condition is reasonable. $SE = 6/\sqrt{25} = 1.2$ and $ME = 2.063899(1.2) = 2.4766788$, so the interval is $42.4 \pm 2.4766788 = (39.9233, 44.8767)$. We are 95% confident that the mean for all 800 transactions is between about 39.923 and 44.877 seconds.

DOK 3 (c) Adjudicate the approaches. Use both endpoints to evaluate the two less-than claims, explain why the cutoff locations matter, distinguish the population mean from individual times, and trace what the rejected interpretation could lead the library to advertise.

Key: Mina is warranted. The whole interval is below 45, so it supports $\mu < 45$. Because 42 is inside, plausible means lie on both sides and $\mu < 42$ is not established. A less-than claim needs the entire interval below its cutoff, not merely some values. The interval estimates the population mean represented by the SRS; it does not contain 95% of individual times. The rejected interpretation could produce unsupported advertisements about both a mean below 42 and individual transaction times.

EXIT

One thing the video changed about my first take: _____